

Name:

Class:

Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.6 Waves

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	

- (a) Write down the equation that links frequency, wave speed and wavelength.

(1)

- (b) The electromagnetic waves used by a Bluetooth wireless connection have a frequency of 2 400 000 000 Hz.

The speed of electromagnetic waves is 300 000 000 m/s.

Calculate the wavelength of the electromagnetic waves.

Wavelength = _____ m

(3)

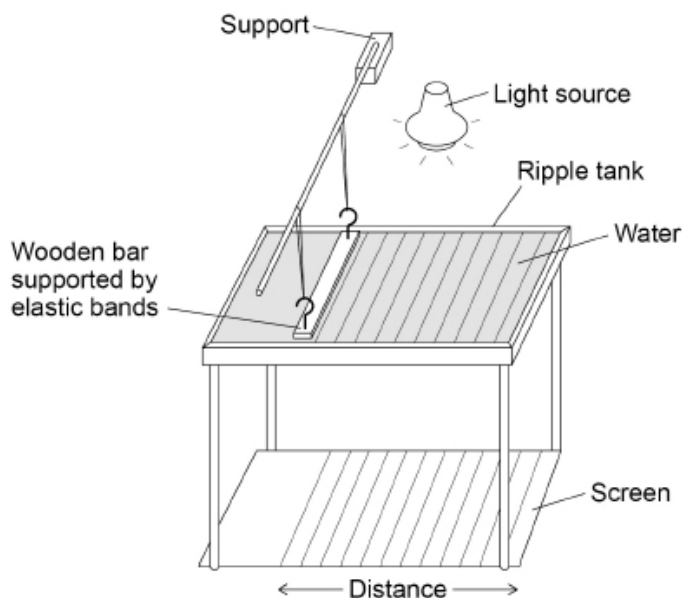
[4 marks]

Figure 1 below shows the equipment a teacher used to determine the speed of a water wave.

The equipment includes:

- A ripple tank filled with water.
- A wooden bar that creates ripples on the surface of the water.
- A light source which causes a shadow of the ripples on the screen.

Figure 1



Describe how equipment in **Figure 1** can be used to measure the wavelength, frequency and speed of a water wave.

02.2

The plastic duck moved up and down as the waves in the water passed.

Figure 2

How does the movement of the plastic duck in **Figure 2** demonstrate that water waves are transverse?

02.3

The teacher measured the maximum height and the minimum height of the plastic duck above the screen as the wave passed.

The teacher repeated his measurements.

Table 1 shows the teacher's measurements.

Table 1

Maximum height in mm	509	513	511
Minimum height in mm	503	498	499

Calculate the mean amplitude of the water wave.

Mean amplitude = _____ mm

[3 marks]

Figure 3 shows the apparatus a student used to investigate the reflection of light by a plane mirror.

The student drew four ray diagrams for each angle of incidence.

The student measured the angle of reflection from each diagram.

Figure 3

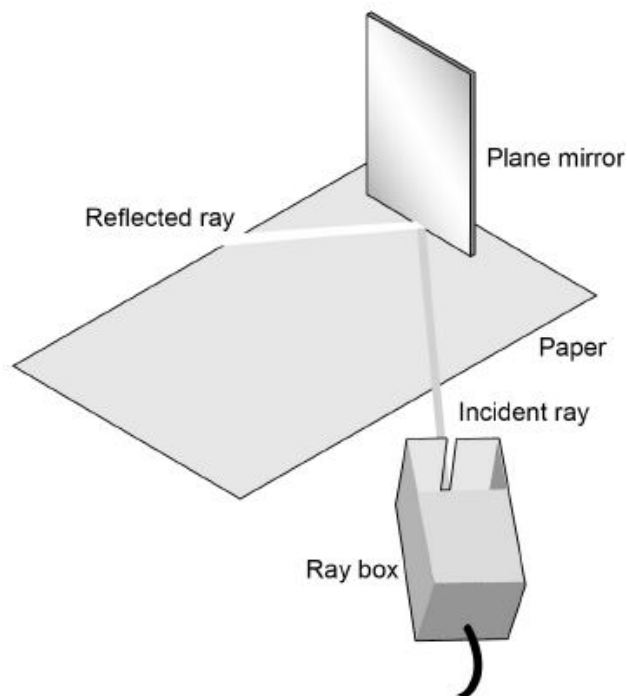


Table 2 gives the student's results.

Table 2

Angle of incidence	Angle of reflection			
	Test 1	Test 2	Test 3	Test 4
20°	19°	22°	20°	19°
30°	31°	28°	32°	30°
40°	42°	40°	43°	41°
50°	56°	49°	53°	46°

Estimate the uncertainty in the angle of reflection when the angle of incidence is 50°.

Show how you determine your estimate.

Uncertainty = $\pm$ _____ °

[2 marks]

03.2

- (a) The student concluded that for a plane mirror, the angle of incidence is equal to the angle of reflection.

Explain whether you agree with this conclusion.

Use examples from the results in the table in your answer.

(2)

- (b) What extra evidence could be collected to support the student's conclusion?

(1)

[3 marks]

03.3

State **one** change the student should make to the apparatus if he wants to use the same method to investigate diffuse reflection.

[1 mark]

04.1

P-waves and S-waves are two types of seismic wave caused by earthquakes.

Which **one** of the statements about P-waves and S-waves is correct?

Tick **one** box.

- ☐ P-waves and S-waves are transverse.
- ☐ P-waves and S-waves are are longitudinal.
- ☐ P-waves are transverse and S-waves are longitudinal.
- ☐ P-waves are longitudinal and S-waves are transverse.

[1 mark]

04.2

Explain why the study of seismic waves provides evidence for the structure of the Earth's core.

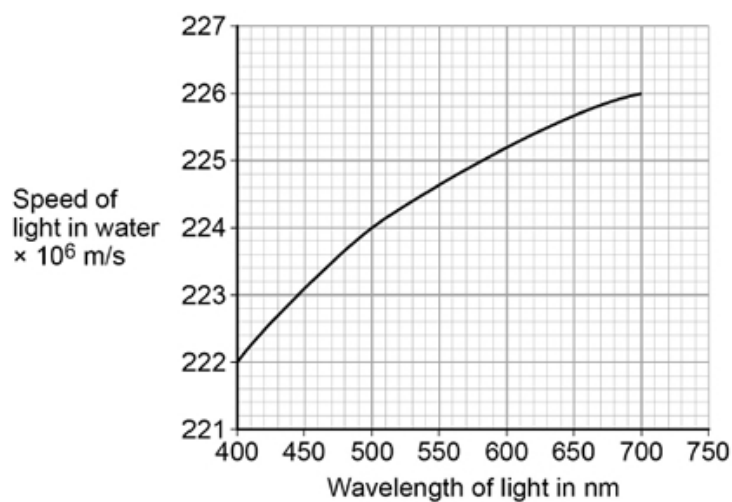
[2 marks]

White light is made up of a continuous spectrum of different wavelengths that all travel at 3×10^8 m/s in air.

Rainbows are produced because different wavelengths of light travel at different speeds in water.

Figure 4 shows the speed of different wavelengths of light in water.

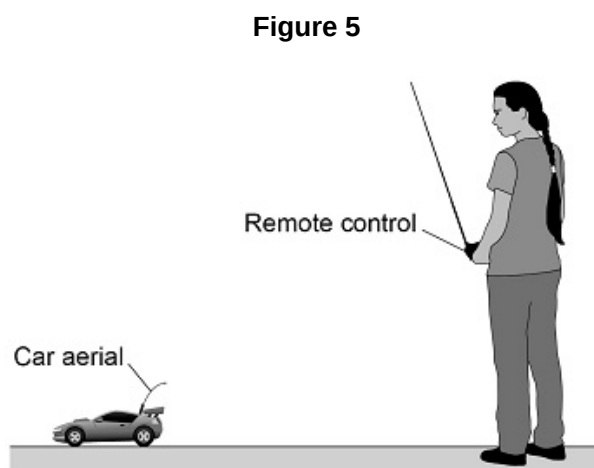
Figure 4



Explain why violet light is refracted the most as it enters water.

[3 marks]

Figure 5 shows a student playing with a remote-controlled car.



The car produces sound waves.

Give **two** ways in which radio waves are different to sound waves.

1.

2.

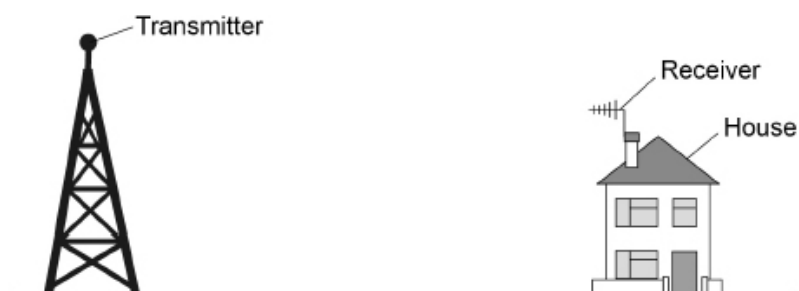
[2 marks]

07

Radio waves form part of the electromagnetic spectrum.

Figure 6 shows one use of radio waves.

Figure 6



Explain how electrical signals in the transmitter produce a signal in the receiver.

[3 marks]

08

Explain how the properties of X-rays make them suitable for the medical imaging of bones.

[2 marks]

X-rays form part of the electromagnetic spectrum.

Radiographers use X-rays to produce images of bones inside the body.

Table 3 shows the effect of exposure to different doses of radiation.

Table 3

Dose in mSv	Effect on the human body
100	slightly increased risk of cancer
1000	5% increased risk of cancer
5000	high risk of death

During an X-ray a person receives a dose of 0.5 mSv.

The radiographer takes many X-ray images each day.

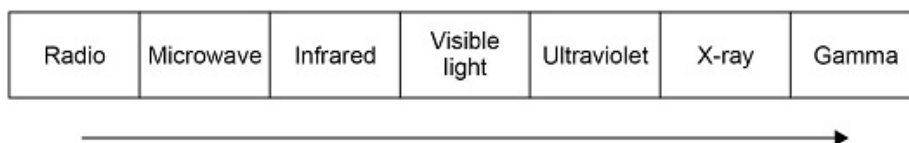
Explain why the radiographer stands behind a protective screen when taking an X-ray image.

[3 marks]

10.1

Figure 7 shows the electromagnetic spectrum.

Figure 7



Which statement is correct for the direction of the arrow?

Tick **one** box.

- ☐ The wavelength decreases and the wave speed in air increases
- ☐ The frequency increases and the wavelength increases
- ☐ The frequency increases and the wave speed in air stays the same
- ☐ The wavelength increases and the wave speed in air increases

[1 mark]

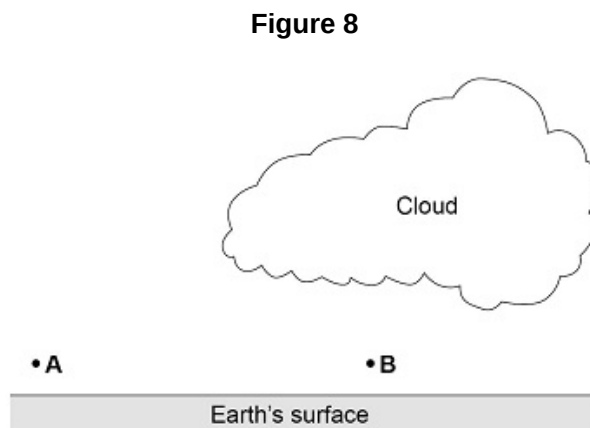
10.2

Albedo is a measure of the amount of solar radiation reflected by an object compared to the total solar radiation incident on the object.

A perfect reflector has an Albedo value of 1.0

A perfect absorber has an Albedo value of 0.0

Figure 8 shows two points, **A** and **B**, just above the Earth's surface.



The average Albedo value of the Earth's surface is 0.3

The Albedo value of thick cloud varies between 0.6 and 0.9

At night the air at point **A** cools faster than the air at point **B**.

Explain why.

[3 marks]